Autonomous Quant Lab

Adversarial Multi-Agent Systems for Scientific Alpha Discovery in the S&P 500 Universe

The Research Objective

Eliminating human bias and overfitting through agentic scientific inquiry.

Solving the Alpha Decay Problem

The Challenge

Traditional quantitative research is slow, manual, and prone to "backtest overfitting." As patterns are discovered, they decay, requiring a perpetual discovery loop.

Our solution utilizes **Large Action Models (LAMs)** to autonomously hypothesize, code, and stress-test strategies at scale.

Core Innovation

Transitioning AI from a passive assistant to an autonomous researcher that adversarialy challenges its own logical conclusions.

Data Infrastructure

The Numerical and Textual foundation for deterministic backtesting.

Numerical Engine: S&P 500 OHLCV

Stock Market Chart Screen

- **Historical Horizon:** 10+ years of daily pricing data.
- **Granularity:** Open, High, Low, Close, and Volume (OHLCV).
- **Integrity:** Decoupled local data lake to avoid API latency and rate-limiting.
- **Bias Control:** Point-in-time data architecture to mitigate survivorship bias.

Textual Engine: SEC 10-K Filings

- **Source:** SEC EDGAR programmatically extracted via *edgartools*.
- **Semantic Input:** Raw 10-K and 10-Q Annual and Quarterly reports.
- **Analysis Items:** Item 1A (Risk Factors) and Item 7 (MD&A).
- **Objective:** Discovery of fundamental anomalies that price-data alone cannot reveal.

The Multi-Agent Architecture

A closed-loop system of checks and balances for robust strategy discovery.

The Cognitive Research Loop

The Creator

Synthesizes economic hypotheses and writes executable Backtrader Python code.

The Critic

Adversarial risk officer that injects slippage, latency, and friction to break strategies.

The Judge

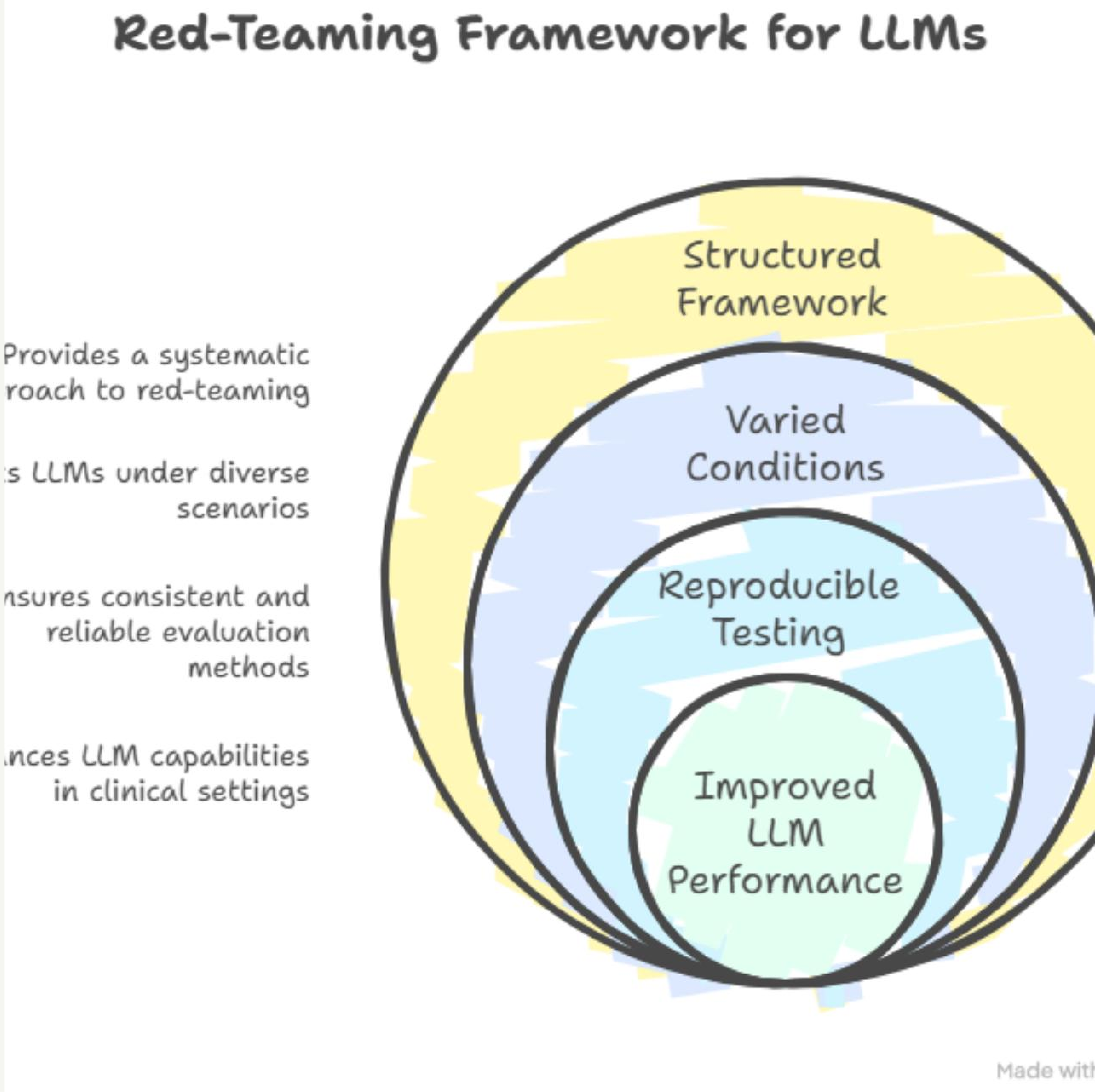
Audits semantic logic and manages ChromaDB memory to ensure rational discovery.

Adversarial Stress Testing

The Critic Agent employs a **Red Team** methodology to expose fragile Alpha.

By forcing execution in sub-optimal environments—simulating 500ms latency and 0.1% slippage—we filter out 80% of strategies that rely on "perfect world" simulations.

This ensures that only industrially robust strategies reach the final portfolio vault.



6-Phase Implementation Roadmap

P1: Data
Numerical & Textual Ingestion

P2: Sandbox
Backtrader Simulation Engine

P3: Memory
ChromaDB Vector Ledger

P4: Brain
Creator Agent Orchestration

P5: Loop
Critic & Judge Adversarial Check

P6: Alpha
24/7 Research Automation

Evaluation & Paper Validation

85%

Code Fix Rate

Autonomous self-correction of syntax errors via sandbox feedback.

1.2+

Target Sharpe

Minimum risk-adjusted return threshold for strategy validation.

Questions?

Autonomous Quant Lab: Towards Scientific Alpha Discovery

Contact: quant_lab@research.edu

Image Sources

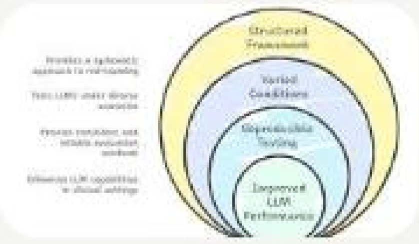
https://images.stockcake.com/public/f/3/2/f32b2961-056c-4c0c-b907-5b52110b7884_large/trading-desk-glow-stockcake.jpg

Source: stockcake.com

Thumbnail for www.quantillium.com

https://cdn.prod.website-files.com/67484cd9cb3e42d3ca6c4c21/6919b7d08db2cf6fcdb247b0_Annual%20Report%20vs%20Form%2010%20K.%20What%20you%20need%20to%20know.webp

Source: www.quantillium.com



https://insights2techinfo.com/wp-content/uploads/2025/12/a-diagram-of-a-diagram-ai-generated-content-may-b.png

Source: insights2techinfo.com